

VisionAid – Abstract

Independent mobility in urban environments remains one of the most significant daily challenges faced by blind and visually impaired individuals. However, many solutions share a critical limitation: they are unable to detect and signal the presence of fast-approaching hazards in real time. This gap in existing assistive technology represents both a safety risk and a barrier to full independence for visually impaired users.

VisionAid is a project developed to address this limitation through the integration of artificial intelligence, mobile development, and a structured support infrastructure. The system consists of two main components: a mobile application and a web-based support ticketing platform.

The mobile application is built using Flutter, ensuring cross-platform compatibility with both Android and iOS devices. It continuously captures live camera frames and transmits them to a remote server running a YOLOv8 object detection model. This server-side inference architecture was chosen deliberately: by offloading the computational workload to a remote machine, the system avoids the hardware constraints of mobile devices and allows the use of a more capable and accurate detection model. The server processes each frame and identifies the following urban hazard categories: cars, motorcycles, buses, trucks, and bicycles. Detection results are sent back to the application, which overlays bounding boxes on the live camera feed and simultaneously triggers two distinct alert channels: spoken warnings via text-to-speech synthesis and haptic feedback through device vibration. This multimodal alert system ensures that users receive instant, unambiguous notifications regardless of surrounding noise or environmental conditions.

The second component is a web-based support ticketing system designed to provide users with reliable access to technical assistance. Through the app, users can submit reports related to issues. Support staff can then manage and respond to incoming tickets through the management interface, ensuring a structured and timely response process.

The results of the project demonstrate that combining server-side AI inference with a carefully designed mobile interface and an organized support system can produce a concrete, accessible, and scalable tool for improving the safety and independence of visually impaired users in everyday urban life. VisionAid serves as a practical example of how modern technology, when developed with accessibility at its core, can make a measurable difference in people's lives.